Name: ______________________________ Date: _________________ Class: _______________

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Sample Collection (Suspended in Water)

We want to get our samples of microplastics as large and pure as possible and within a shorter collection time.

Movement/Maneuverability

We want to have a neutrally buoyant ROV that can navigate to any reasonable spot. Our axes of motion that should work are forwards, backwards, up, down, and yaw.

In addition to basic motion, we want our ROV to be stable, maneuverable and fast. This will be dependent on motor placement and buoyancy.

Reusability

We want to minimize how our ROV decays after use; we want it to function as well as possible after heavy usage and minimal maintenance.

Additional Data Collection

We must be able to collect sample data outside of the ROV such as water speed, surface temperature, time spent in water, and possibly video feeds.

Sediment Sample Collection

The sediment sample collector will need to be able to collect a large sample of sea floor sediment, hold on to it, and transport it back to the surface in the ROV.

Intuitive Controls and Ease of Operation

Ease of operation plays into the goal of maneuverability, but also includes controls, electronics, video feeds, and overall operator friendliness.

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Option 1, /

1. Any equilibrium solutions (if there are no equilibrium solutions, state why not)
    1. There are no equilibrium solutions. If there were that would imply working values for for all values of x, for which y would have to be infinitely large.
2. The regions of increasing and decreasing slopes show your work.
    1. I found that for the slope to be positive the signs of x and y would have to be different, (the upper left and bottom right quadrants), and for the slope to be negative the signs would have to be the same (the upper right and bottom left quadrants).
3. Where are the changes in concavity, show your work.
    1. The work is shown in the attached image, but I found that and since the numerator is always positive, the solutions are concave up when $y < 0$ and concave down when $y > 0$.
4. Draw two solution curves, identify what the initial conditions are and make sure that your curve shows what occurs as it gets large.
    1. I attached my drawings below. They're on the upper part of the second page. The initial conditions chosen were $y(0) = 1$ and $y(0) = 2$. I did also find the domain of x, since it's not defined past a certain point. As for very large values of y', that occurs for very small values of y, and very large values of x, which makes sense since the ellipses point straight up when crossing the x axis. This also means there are slopes of zero across the y axis, which also makes a lot of sense.

Here is my work and the drawing I did for the slope field.

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